

Exercise sheet 8

This assignment is due on March 20th at noon, to be submitted electronically via Blackboard.

1. Let $\text{Ai}(x), x \in \mathbb{R}$ denote the Airy function defined as

$$\text{Ai}(x) := \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt,$$

and which solves the differential equation

$$\frac{d^2}{dx^2} \text{Ai}(x) = x \text{Ai}(x)$$

such that

$$\text{Ai}(x) = \frac{1}{2\sqrt{\pi}} x^{-\frac{1}{4}} e^{-\frac{2}{3}x^{\frac{3}{2}}} \left(1 + O(x^{-\frac{3}{2}})\right), \quad x \rightarrow +\infty.$$

- (a) Show that for all $x, y \in \mathbb{R}$,

$$(x - y) \int_0^\infty \text{Ai}(x + u) \text{Ai}(u + y) du = \text{Ai}(x) \frac{d}{dy} \text{Ai}(y) - \text{Ai}(y) \frac{d}{dx} \text{Ai}(x).$$

- (b) Hence show that

$$\int_0^\infty \text{Ai}(x + u) \text{Ai}(x + u) du = \left(\frac{d}{dx} \text{Ai}(x)\right)^2 - x \text{Ai}^2(x).$$

2. Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{E}[X_1] = \mu > 0$ and $\mathbb{E}[X_1^2] = t > 0$. Show that

- (a) $\text{Var}(X_1) = t - \mu^2$.
- (b) Define $S_n = \frac{1}{n} \sum_{j=1}^n X_j$. Show that $\mathbb{E}[S_n]$ is order one in n (i.e. it is constant as $n \rightarrow \infty$).
- (c) Show that the standard deviation of S_n is of size $1/\sqrt{n}$. (Thus the fluctuations around the order one expected value are of size $1/\sqrt{n}$.)
- (d) Compare this level of fluctuation to the scaling (16.38) of the maximal GUE eigenvalue in the notes for the maximal eigenvalue. Would we expect to see a maximal eigenvalue closer or further away from its order-one mean $\sqrt{2}$ than for S_n ?

3. Suppose m_k are the moments of the semicircle law, i.e. for $\sigma_t(x) := \frac{1}{2\pi t} \sqrt{(4t - x^2)_+}$ and any $k = 0, 1, 2, \dots$,

$$m_k = \int_{\mathbb{R}} x^k \sigma_1(x) dx.$$

Show that for $k = 0, 1, 2, \dots$

(a) and for any $t > 0$,

$$\int_{\mathbb{R}} x^k \sigma_t(x) dx = t^{\frac{k}{2}} m_k,$$

(b) $m_0 = 1$,

(c) $m_{2k-1} = 0$,

(d) $m_{2k} = \frac{2(2k-1)}{k+1} m_{2k-2}$.

(e) Hence show that

$$m_{2k} = \frac{1}{k+1} \binom{2k}{k}, \quad k \in \mathbb{Z}_{\geq 0}.$$

4. (Non-compulsory but encouraged!) In your favourite suitable coding language (e.g. Python)

- a) build a real Wigner matrix M_N (you can pick your distributions!) of some large-ish size (e.g $N = 50$)
- b) compute the eigenvalues (hint: using that M_N is symmetric can speed up this computation)
- c) plot the resulting histogram.

Hopefully you will see the semicircle! What happens if you change the variance of your off-axis entries? What happens if you repeat the experiment for a large number of matrices from your ensemble? What happens if you square all your eigenvalues and then plot the histogram?